

Polar Unit Test (Practice C)

Calculator

You have the entire period.

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test. _____

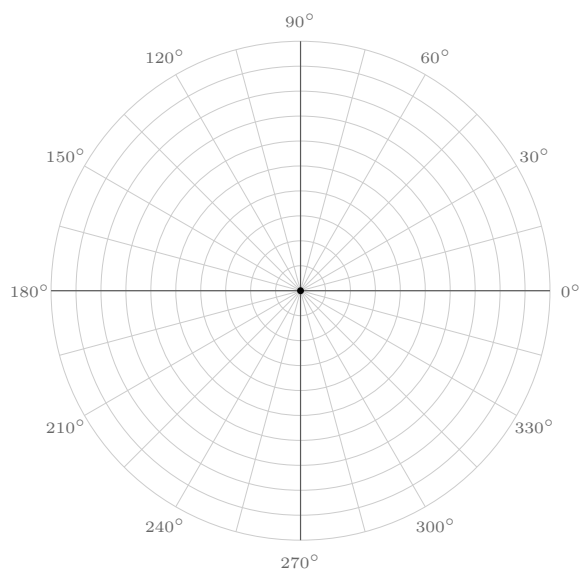
SHOW ALL WORK FOR FULL CREDIT.

Signature _____

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. Show your set up (what you are putting into the calculator). (3 points each)

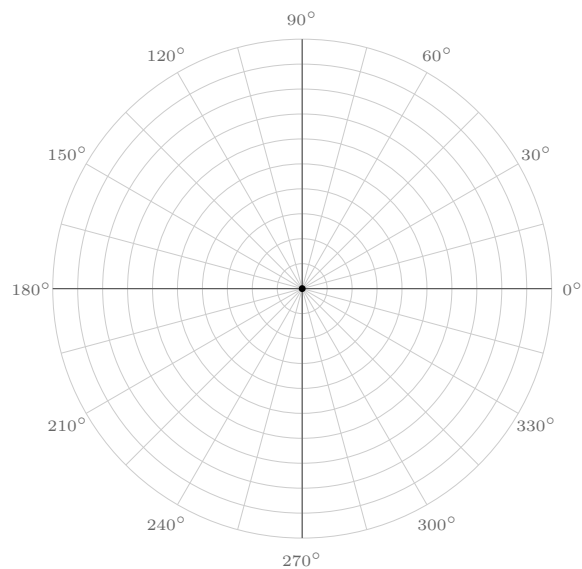
Word bank: circle cardioid limaçon rose curve lemniscate

1. $r = 4 + 2 \cos \theta$



Type: _____

2. $r = 5 \cos 2\theta$



Type: _____

Polar Unit Test (Practice C)

Non-Calculator

SHOW ALL WORK FOR FULL CREDIT.

Simplify. (3 points) 3. $\frac{\sqrt{6}}{\sqrt{-3} \cdot \sqrt{-8}}$	Simplify. (2 points) 4. $\frac{-8 + \sqrt{-48}}{12}$
Perform the indicated operation. Write the answer in rectangular form. (2 points) 5. i^{75}	Perform the indicated operation. Write the answer in rectangular form. (2 points) 6. $(-2 + 5i) - (4 - 3i) - (-7 + i)$
Perform the indicated operation. Write the answer in rectangular form. (3 points) 7. $\frac{\sqrt{18} + i\sqrt{18}}{\sqrt{2} - i\sqrt{2}}$	Perform the indicated operation. Write the answer in rectangular form. (3 points) 8. $3i(2 - i)(1 + 2i)$
Solve each equation over the set of complex numbers. (3 points) 9. $x^2 + 13 = 4x$	

Convert to trig form. (3 points)

10. $-2\sqrt{3} + 2i$

Convert to rectangular form. (3 points)

11. $8 \operatorname{cis} \frac{4\pi}{3}$

Using $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$ is the following in the Julia set? Show ALL work. (3 points)

12. $z = 0 + i$

YES / NO

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

13. $\frac{6\sqrt{2}(\cos 75^\circ + i \sin 75^\circ)}{3(\cos 300^\circ + i \sin 300^\circ)}$

14. $\left[3 \operatorname{cis} \frac{5\pi}{4}\right] \left[2 \operatorname{cis} \frac{3\pi}{2}\right]$

Using $z, z^2 + z, (z^2 + z)^2 + z, \dots$ is the following number in the Mandelbrot set? Show ALL work. (3 points)

15. $z = -1 + i$

YES / NO

Find each power. Write your answer in rectangular form. (3 pts)

16. $(1 + i\sqrt{3})^7$

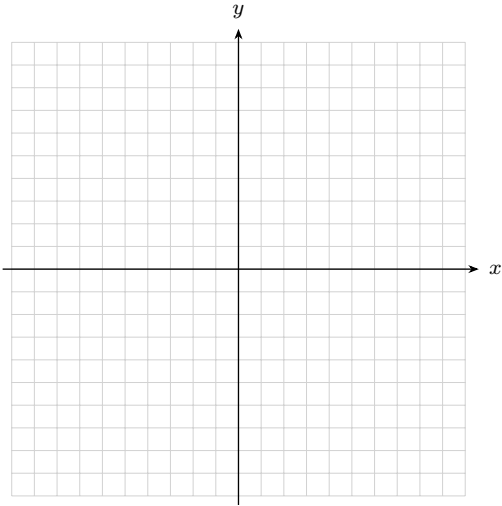
Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17. $x^3 - 27i = 0$

Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points) 18. $(-4, 210^\circ)$	Convert the rectangular coordinates to polar coordinates with $r > 0$ and $0^\circ \leq \theta < 360^\circ$. (3 points) 19. $(-5, 5\sqrt{3})$
Convert the rectangular equation to a polar equation. Solve for r. (3 points) 20. $4x + 5y = 10$	Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points) 21. $(8, 0^\circ)$ and $(2, 270^\circ)$

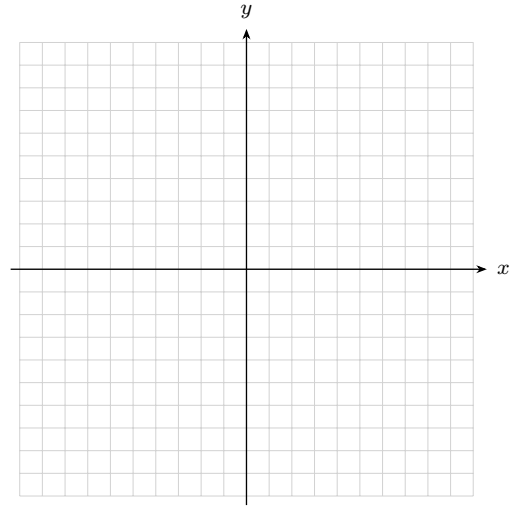
Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

22. $r = 10 \sin \theta$



Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23. $r = \frac{6}{2 \cos \theta + \sin \theta}$



Find the polar coordinates of the points of intersection for the given interval of θ . (4 points)

24. $r = 4 + 4 \sin \theta$, $r = 6$
 $0 \leq \theta < 2\pi$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25. $r = 7 \cos \theta$

Type: _____

ANSWER KEY – Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Pace check: about 50 minutes for a strong pass. Calculator part about 5, #3–12 about 15, #13–17 about 13, #18–25 about 15. That leaves about 30 minutes of the period to check your work.

Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

1. $r = 4 + 2 \cos \theta$

WORK

Polar mode: $r1(\theta) = 4 + 2 \cos \theta$

$\theta = 0^\circ: r = 6; 90^\circ, 270^\circ: r = 4; 180^\circ: r = 2$

$\frac{a}{b} = \frac{4}{2} = 2 \neq 1$

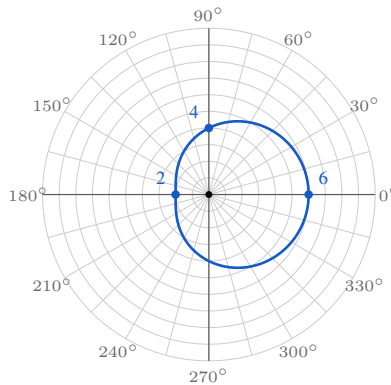
limaçon

WHY

Degree mode needs θ from 0 to 360 (radians: 0 to 2π).

Graph Trace reads r at each angle.

Not 1, so not a cardioid. Never reaches the pole.



Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

2. $r = 5 \cos 2\theta$

WORK

Polar mode: $r1(\theta) = 5 \cos(2\theta)$

$n = 2$ (even): $2n = 4$ petals

tips at $0^\circ, 90^\circ, 180^\circ, 270^\circ$, each 5 long

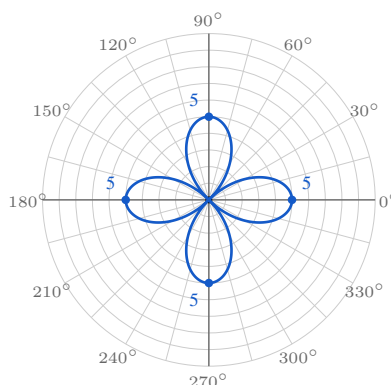
rose curve

WHY

Parentheses around 2θ .

n even doubles.

$\cos$ puts a petal on the x -axis.



ANSWER KEY – Non-Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Simplify. (3 points)

3. $\frac{\sqrt{6}}{\sqrt{-3} \cdot \sqrt{-8}}$

WORK

$$\begin{aligned}\frac{\sqrt{6}}{i\sqrt{3} \cdot i\sqrt{8}} &= \frac{\sqrt{6}}{i^2\sqrt{24}} \\ &= \frac{\sqrt{6}}{-2\sqrt{6}} = \boxed{-\frac{1}{2}}\end{aligned}$$

Trap: $\sqrt{-3} \cdot \sqrt{-8}$ is not $\sqrt{24}$. Pull out i first or the sign flips.

WHY

Pull out i first.

$\sqrt{24} = 2\sqrt{6}$; the $\sqrt{6}$ cancels.

Simplify. (2 points)

4. $\frac{-8 + \sqrt{-48}}{12}$

WORK

$$\begin{aligned}\sqrt{-48} &= i\sqrt{16 \cdot 3} = 4i\sqrt{3} \\ \frac{-8 + 4i\sqrt{3}}{12} &= \boxed{-\frac{2}{3} + \frac{\sqrt{3}}{3}i}\end{aligned}$$

WHY

Split, reduce each part by 4.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

5. i^{75}

WORK

$$75 \div 4 = 18 \text{ R } 3$$

$$i^3 = \boxed{0 - i}$$

WHY

Only the remainder matters.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

6. $(-2 + 5i) - (4 - 3i) - (-7 + i)$

WORK

$$-2 + 5i - 4 + 3i + 7 - i$$

$$\text{real: } -2 - 4 + 7 = 1; \text{ imag: } 5 + 3 - 1 = 7$$

$$\boxed{1 + 7i}$$

WHY

Each minus flips both parts.

Trap: $-(-7 + i)$ is $+7 - i$: both signs flip.

Perform the indicated operation. Write the answer in rectangular form. (3 points)

7. $\frac{\sqrt{18} + i\sqrt{18}}{\sqrt{2} - i\sqrt{2}}$

WORK

$$\cdot \frac{\sqrt{2} + i\sqrt{2}}{\sqrt{2} + i\sqrt{2}}$$

top: $\sqrt{36} + i\sqrt{36} + i\sqrt{36} + i^2\sqrt{36} = 6 + 12i - 6 = 12i$

bottom: $2 + 2 = 4$

$$\frac{12i}{4} = \boxed{0 + 3i}$$

WHY

Conjugate of the bottom.

$$\sqrt{18}\sqrt{2} = 6.$$

$$a^2 + b^2.$$

Perform the indicated operation. Write the answer in rectangular form. (3 points)

8. $3i(2 - i)(1 + 2i)$

WORK

$$(2 - i)(1 + 2i) = 2 + 4i - i - 2i^2 = 4 + 3i$$

$$3i(4 + 3i) = 12i + 9i^2 = \boxed{-9 + 12i}$$

WHY

FOIL first two.

$$i^2 = -1.$$

Solve each equation over the set of complex numbers. (3 points)

9. $x^2 + 13 = 4x$

WORK

$$x^2 - 4x + 13 = 0$$

$$x = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2}$$

$$\boxed{x = 2 \pm 3i}$$

WHY

Set equal to 0.

Quadratic formula.

Convert to trig form. (3 points)

10. $-2\sqrt{3} + 2i$

WORK

$$r = \sqrt{12 + 4} = 4$$

$$\tan^{-1}\left(\frac{2}{-2\sqrt{3}}\right) \rightarrow \text{ref } 30^\circ, \text{ QII: } 150^\circ$$

$$\boxed{4(\cos 150^\circ + i \sin 150^\circ)}$$

WHY

$$x < 0, y > 0.$$

Convert to rectangular form. (3 points)

11. $8 \operatorname{cis} \frac{4\pi}{3}$

WORK

$$8\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$$

$$\boxed{-4 - 4\sqrt{3}i}$$

WHY

$$\frac{4\pi}{3} \text{ is QIII.}$$

Using $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... is the following in the Julia set? Show ALL work. (3 points)

12. $z = 0 + i$

YES / NO

WORK

$$i^2 - 1 = -2, |-2| = 2$$

$$(-2)^2 - 1 = 3, |3| = 3 > 2$$

NO

WHY

2 does not exceed 2; keep going.

Exceeds 2.

Trap: Stopping at $|-2| = 2$ and saying yes. Equal to 2 is not exceeding it; take one more step.

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

13. $\frac{6\sqrt{2}(\cos 75^\circ + i \sin 75^\circ)}{3(\cos 300^\circ + i \sin 300^\circ)}$

WORK

$$2\sqrt{2} \operatorname{cis}(75^\circ - 300^\circ) = 2\sqrt{2} \operatorname{cis}(-225^\circ) = 2\sqrt{2} \operatorname{cis} 135^\circ$$

$$2\sqrt{2} \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right) = \boxed{-2 + 2i}$$

WHY

Divide r 's, subtract angles, add 360° .

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

14. $\left[3 \operatorname{cis} \frac{5\pi}{4} \right] \left[2 \operatorname{cis} \frac{3\pi}{2} \right]$

WORK

$$6 \operatorname{cis} \left(\frac{5\pi}{4} + \frac{6\pi}{4} \right) = 6 \operatorname{cis} \frac{11\pi}{4}$$

$$\frac{11\pi}{4} - 2\pi = \frac{3\pi}{4}$$

$$6 \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right) = \boxed{-3\sqrt{2} + 3\sqrt{2}i}$$

WHY

Multiply r 's, add angles.

Subtract 2π , not π .

Using z , $z^2 + z$, $(z^2 + z)^2 + z$, ... is the following number in the Mandelbrot set? Show ALL work. (3 points)

15. $z = -1 + i$

YES / NO

WORK

$$|z| = \sqrt{2}$$

$$(-1 + i)^2 + z = -2i - 1 + i = -1 - i, \sqrt{2}$$

$$(-1 - i)^2 + z = 2i - 1 + i = -1 + 3i, \sqrt{10} > 2$$

NO

WHY

Start.

$$(-1 + i)^2 = 1 - 2i + i^2 = -2i.$$

Exceeds 2.

Find each power. Write your answer in rectangular form. (3 pts)

16. $(1 + i\sqrt{3})^7$

WORK

$$r = 2, \theta = 60^\circ$$

$$2^7 \operatorname{cis} 420^\circ = 128 \operatorname{cis} 60^\circ$$

$$128 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = \boxed{64 + 64\sqrt{3}i}$$

WHY

QI.

De Moivre; $420 - 360 = 60$.

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17. $x^3 - 27i = 0$

WORK

$$x^3 = 27i = 27 \operatorname{cis} 90^\circ$$

$$\sqrt[3]{27} = 3, \frac{90^\circ + 360^\circ k}{3} = 30^\circ + 120^\circ k$$

$$\boxed{3 \operatorname{cis} 30^\circ, 3 \operatorname{cis} 150^\circ, 3 \operatorname{cis} 270^\circ}$$

WHY

Isolate; $27i$ points straight up.

$$k = 0, 1, 2.$$

Trig form asked: stop.

Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points)

18. $(-4, 210^\circ)$

WORK

$$x = -4 \cos 210^\circ = -4 \left(-\frac{\sqrt{3}}{2} \right) = 2\sqrt{3}$$

$$y = -4 \sin 210^\circ = -4 \left(-\frac{1}{2} \right) = 2$$

$$\boxed{(2\sqrt{3}, 2)}$$

WHY

Keep r 's sign.

Negative r : opposite of QIII is QI.

Convert the rectangular coordinates to polar coordinates with $r > 0$ and $0^\circ \leq \theta < 360^\circ$. (3 points)

19. $(-5, 5\sqrt{3})$

WORK

$$r = \sqrt{25 + 75} = 10$$

$$\text{ref } 60^\circ, \text{ QII: } 120^\circ$$

$$\boxed{(10, 120^\circ)}$$

WHY

$$x < 0, y > 0.$$

Trap: $\tan^{-1}(-\sqrt{3}) = -60^\circ$ is the wrong quadrant. Check signs of x and y .

Convert the rectangular equation to a polar equation. Solve for r . (3 points)

20. $4x + 5y = 10$

WORK

$$4r \cos \theta + 5r \sin \theta = 10$$

$$\boxed{r = \frac{10}{4 \cos \theta + 5 \sin \theta}}$$

WHY

Substitute.

Factor r , divide.

Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points)

21. $(8, 0^\circ)$ and $(2, 270^\circ)$

WORK

$$(8, 0) \text{ and } (0, -2)$$

$$m = \frac{-2 - 0}{0 - 8} = \frac{1}{4}, y = \frac{1}{4}x - 2$$

$$x - 4y = 8$$

$$\boxed{r = \frac{8}{\cos \theta - 4 \sin \theta}}$$

WHY

270° points down.

Times 4, standard form.

Substitute, factor, divide.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

22. $r = 10 \sin \theta$

WORK

$$r^2 = 10r \sin \theta, \quad x^2 + y^2 = 10y$$

$$x^2 + y^2 - 10y + 25 = 25$$

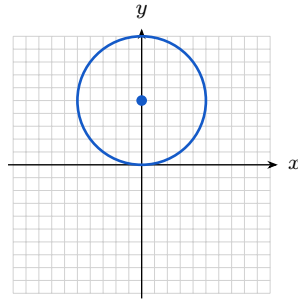
$$x^2 + (y - 5)^2 = 25$$

WHY

Multiply by r , substitute.

Complete the square.

Circle, center $(0, 5)$, radius 5.



Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23. $r = \frac{6}{2 \cos \theta + \sin \theta}$

WORK

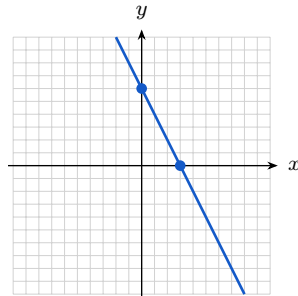
$$2r \cos \theta + r \sin \theta = 6, \quad 2x + y = 6$$

$$y = -2x + 6$$

WHY

Clear fraction, substitute.

Slope -2 , intercepts $(3, 0)$, $(0, 6)$.



Find the polar coordinates of the points of intersection for the given interval of θ . (4 points)

24. $r = 4 + 4 \sin \theta, \quad r = 6$
 $0 \leq \theta < 2\pi$

WORK

$$4 + 4 \sin \theta = 6, \quad \sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \quad \frac{5\pi}{6}$$

$$\left(6, \frac{\pi}{6}\right), \left(6, \frac{5\pi}{6}\right)$$

WHY

Set $r = r$.

Sine positive: QI, QII.

$r = 6$ never hits the pole.

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25. $r = 7 \cos \theta$

WORK

circle

Trap: $r = a \cos \theta$ is a circle, not a 1-petal rose.

WHY

$n = 1$ is not a rose: $x^2 + y^2 = 7x$.